Q1.What is data cleaning ,and why is important in data analysis?

Ans- Data cleaning ,also called data cleansing or data scrubbing is the process of detecting, correcting, or removing errors and inconsistencies in data to ensure it is accurate, complete and reliable for analysis.in simple terms , it's like tidying up your data so it can give meaningful and trustworthy.

- What are the potential consequences of analyzing unclean or messy data?

Ans- **Analyzing unclean or messy data can lead to a range of negative consequences, especially in decision-making and research. Key potential consequences include:**

1. **Incorrect conclusions and decisions**
   Errors, duplicates, missing values, or inconsistencies can distort results, leading to false insights and poor strategic or operational decisions.
2. **Biased or misleading results**
   Unclean data can introduce systematic bias, causing certain groups, trends, or outcomes to be over- or under-represented.
3. **Reduced model performance**
   In analytics and machine learning, messy data can significantly lower model accuracy, reliability, and generalizability.
4. **Loss of credibility and trust**
   Stakeholders may lose confidence in analyses or reports if results are inconsistent, contradictory, or later proven wrong.
5. **Wasted time and resources**
   Acting on flawed insights can lead to inefficient processes, financial losses, or the need to redo analyses.
6. **Increased error propagation**
   Small data issues can compound as data flows through systems, leading to larger downstream problems.
7. **Compliance and legal risks**
   In regulated industries, inaccurate data can result in reporting errors, violations, or legal penalties.
8. **Difficulty in reproducibility**
   Messy data makes it hard for others to replicate or validate findings, weakening the reliability of the analysis.

In short: unclean data undermines accuracy, trust, and value—making data cleaning a critical step before any meaningful analysis.

- Explain the common steps involved in cleaning and organizing data.

Ans- 1.**Data inspection and understanding**
2.**Removing duplicate data**.

3.**Handling missing values**
4.Correcting errors and inconsistencies

5. Standardizing data formats
6.Validating data accuracy
7.Handling outliers.

8.Organizing and structuring data

9.Documenting changes

Q2.How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach.

Employee	Department	Salary
Sonu	IT	4000
Pranav	HR	5000
Rahul	IT	2500

Ans-

Employee	Department	Salary
Pranav	HR	5000
Sonu	IT	4000
Rahul	IT	2500

Q3. Explain the use of text functions such as TRIM , LEFT, RIGHT, MID, and CONCAT in data cleaning

Ans- **1. TRIM**

Purpose:

- Removes all extra spaces from text **except single spaces between words**.
- Very useful when data is imported from other sources and contains extra spaces that can break sorting, matching, or formulas.

Use in Data Cleaning:

- Remove leading/trailing spaces in names, addresses, IDs, etc.
- Ensures accurate matches when comparing text or joining tables.

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2. LEFT

Purpose:

- Extracts a specific number of characters **from the start** (left) of a text string.

Syntax: LEFT(text, num_chars)

Use in Data Cleaning:

- Extract first few characters of codes or IDs.
- Standardize abbreviations (e.g., first 3 letters of department codes).

3. RIGHT

- **Purpose:**
- Extracts a specific number of characters **from the end** (right) of a text string.
- **Syntax:** RIGHT(text, num_chars)

Use in Data Cleaning:

- Extract year, suffix, or code from the end of a string.
- Useful for IDs, product codes, or dates in text format.

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4. MID

Purpose:

- Extracts characters **from the middle** of a text string.

Syntax: MID(text, start_num, num_chars)

Use in Data Cleaning:

- Extract middle portions of codes, names, or references.
- Split data without using text-to-columns.

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5. CONCAT (or CONCATENATE)

Purpose:

- Combines two or more text strings into one.

Syntax: CONCAT(text1, text2, …)

Use in Data Cleaning:

- Merge first name and last name into full name.
- Combine address parts, IDs, or codes.
- Create standardized labels or identifiers.

Q4.What is the role of date functions like TODAY in managing datasets?

Ans-Role of Date Functions in Managing Datasets

Date functions in Excel (or Google Sheets) are crucial for analyzing, tracking, and managing time-sensitive data. One common function is **TODAY()**.

1. TODAY() Function

Purpose:

- Returns the **current date** (automatically updates every day).
- Does **not require any arguments**.

Syntax: =TODAY()

Example: =IF(DueDate < TODAY(), "Overdue", "On Time")

This automatically marks tasks based on the current date.

Q5.Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10.

a.Configure an input message that appears when a user selects a cell in the "Quantity" column explaining: "Please enter a whole number between 1 and 10."

b.Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing: "Invalid input! The quantity must be a whole number between 1 and 10."

Ans- **a.**Select all Quantity cells (e.g., D2:D10).

Go to Input Message tab:

*Title: Quantity Entry

*Message: “Please enter a whole number between 1 and 10.”

b. Go to Error Alert tab:

- Style: Stop
- Title: Invalid Quantity
- Message: “Invalid input! The quantity must be a whole number between 1 and 10.”

- When a user selects a Quantity cell → input message appears.
- If user enters `<1`, `>10`, or non-integer → error alert pops up.
- Valid entries `1–10` are accepted.

Q6.U nderstand and apply fundamental text functions like LEFT, RIGHT, MID, and LEN.

*Extract the first 5 characters from the string "ExcelTipsAreGreat" using the LEFT function.
*Extract the last 4 characters from "DataAnalysis.xlsx" using the RIGHT function.

*Extract the substring "Tips" from "ExcelTipsAreGreat" using the MID function.

* Count the total number of characters in the string "Hello World!" using the LEN function.
*Create a formula to extract the middle 6 characters from "12345-67890-ABCDE".

Ans- * Extract the first 5 characters using LEFT

Text: ExcelTipsAreGreat

Formula:

=LEFT("ExcelTipsAreGreat", 5)

* Extract the last 4 characters using RIGHT

Text: DataAnalysis.xlsx

Formula:

=RIGHT("DataAnalysis.xlsx", 4)

*** Extract the substring "Tips" using MID**

Text: ExcelTipsAreGreat

Formula:

=MID("ExcelTipsAreGreat", 6, 4)

*** Count total characters using LEN**

Text: Hello World!

Formula:

=LEN("Hello World!")

***Extract the middle 6 characters**

Text: 12345-67890-ABCDE

Formula:

=MID("12345-67890-ABCDE", 7, 6)

Q7. Understand how to combine text using CONCAT, TEXTJOIN, and the & operator.

a.Use CONCAT to combine "Hello" and "World" with a space in between.

b.Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN.

c.Use the & operator to create the string "2025: Excel Functions" by combining "2025", ": ", and "Excel Functions".

d.Create a comma-separated list from the range A1:A5 using TEXTJOIN

e.Combine first names in column A with last names in column B to create full names in column C.

Ans- a.=CONCAT("Hello"," ","World")

b. =TEXTJOIN(", ",TRUE,"Apple","Banana","Cherry")

c. ="2025"&": "&"Excel Functions"

d. =TEXTJOIN(", ",TRUE,A1:A5)

e. Formula (in C1):

=CONCAT(A1," ",B1)

(or using & operator)

=A1&" "&B1
Full names appear in Column C (e.g., *John Smith*).

Q8.Understanding TODAY() and NOW()

a. What is the difference between TODAY() and NOW() in Excel? Provide an example of when you would use each function.

b. If cell A1 contains the date 2025-06-10, write a formula using TODAY() to determine how many days are left until that date

c. Write an Excel formula using NOW() to display the current date and time in the format MM/DD/YYYY HH:MM AM/PM.´

d. If a cell contains =TODAY(), what will happen when the worksheet is reopened the next day? Explain

e. You want to store a static date (today's date) in a cell without it changing every day. What keyboard shortcut should you use?

Ans-a.

Function	What it returns	When to use
TODAY()	Current **date only**	Tracking deadlines, due dates, age or number of days remaining.
NOW()	Current date and time	Time-stamped logs, attendance, system-generated reports.

b. =A1 - TODAY()

c. =TEXT(NOW(),"MM/DD/YYYY HH:MM AM/PM")

d. *The value of =TODAY() automatically updates to the new current date

*Any formulas dependent on it also update.

*This makes reports and calculations dynamic.

e. **Keyboard shortcut:**

- **Ctrl + ;** → Inserts today's date (static)
- **Ctrl + Shift + ;** → Inserts current time (static)